

Digital Electronics - Conversion of Flip-Flops

In previous chapter, we discussed the four flip-flops, namely SR flip-flop, D flip-flop, JK flip-flop & T flip-flop. We can convert one flip-flop into the remaining three flip-flops by including some additional logic. So, there will be total of twelve **flip-flop conversions**.

Follow these **steps** for converting one flip-flop to the other.

- Consider the **characteristic table** of desired flip-flop.
- Fill the excitation values (inputs) of given flip-flop for each combination of present state and next state. The **excitation table** for all flip-flops is shown below.

Present State	Next State	SR Flip-Flop Inputs		D flip-flop input	JK Flip-Flop Inputs		T Flip-Flop Input
Q(t)	Q(t+1)	S	R	D	J	K	T
0	0	0	x	0	0	x	0
0	1	1	0	1	1	x	1
1	0	0	1	0	x	1	1
1	1	x	0	1	x	0	0

Get the **simplified expressions** for each excitation input. If necessary, use Kmaps for simplifying.

Draw the **circuit diagram** of desired flip-flop according to the simplified expressions using given flip-flop and necessary logic gates.

Now, let us convert few flip-flops into other. Follow the same process for remaining flipflop conversions.

SR Flip-Flop to other Flip-Flop Conversions

Following are the three possible conversions of SR flip-flop to other flip-flops.

- SR Flip-Flop to D Flip-Flop
- SR Flip-Flop to JK Flip-Flop
- SR Flip-Flop to T Flip-Flop

SR Flip-Flop to D Flip-Flop Conversion

Here, the given flip-flop is SR flip-flop and the desired flip-flop is D flip-flop. Therefore, consider the following **characteristic table** of D flip-flop.

D Flip-Flop Input	Present State	Next State
D	Q(t)	Q(t + 1)
0	0	0
0	1	0
1	0	1
1	1	1

We know that SR flip-flop has two inputs S & R. So, write down the excitation values of SR flip-flop for each combination of present state and next state values. The following table shows the characteristic table of D flip-flop along with the **excitation inputs** of SR flip-flop.

D Flip-Flop Input	Present State	Next State	SR Flip-Flop Inputs	
D	Q(t)	Q(t + 1)	S	R

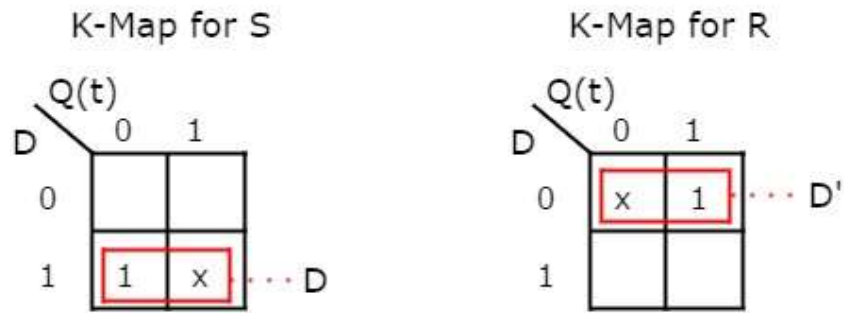
0	0	0	0	x
0	1	0	0	1
1	0	1	1	0
1	1	1	x	0

From the above table, we can write the **Boolean functions** for each input as below.

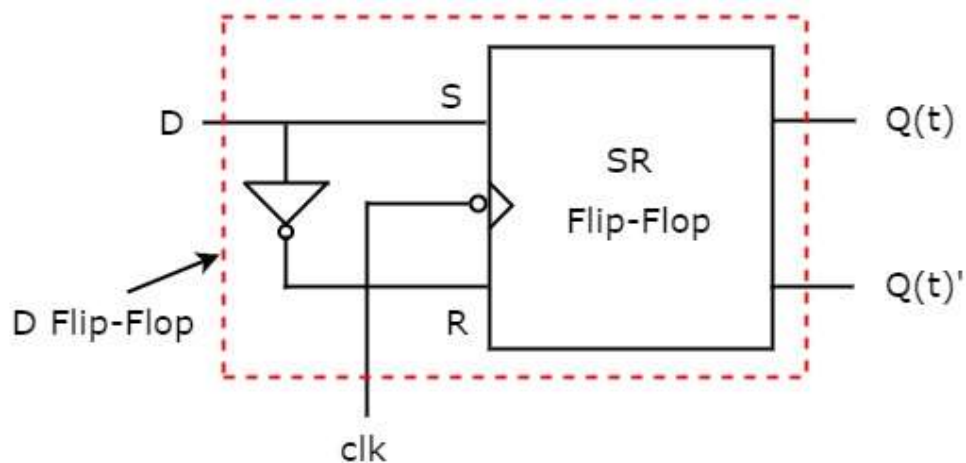
$$S = m_2 + d_3$$

$$R = m_1 + d_0$$

We can use 2 variable K-Maps for getting simplified expressions for these inputs. The **k-Maps** for S & R are shown below.



So, we got $S = D$ & $R = D'$ after simplifying. The **circuit diagram** of D flip-flop is shown in the following figure.



This circuit consists of SR flip-flop and an inverter. This inverter produces an output, which is complement of input, D. So, the overall circuit has single input, D and two outputs $Q(t)$ & $Q(t)'$. Hence, it is a **D flip-flop**. Similarly, you can do other two conversions.

D Flip-Flop to other Flip-Flop Conversions

Following are the three possible conversions of D flip-flop to other flip-flops.

- D Flip-Flop to T Flip-Flop
- D Flip-Flop to SR Flip-Flop
- D Flip-Flop to JK Flip-Flop

D Flip-Flop to T Flip-Flop conversion

Here, the given flip-flop is D flip-flop and the desired flip-flop is T flip-flop. Therefore, consider the following **characteristic table** of T flip-flop.

T Flip-Flop Input	Present State	Next State
T	Q(t)	Q(t + 1)
0	0	0
0	1	1
1	0	1
1	1	0

We know that D flip-flop has single input D. So, write down the excitation values of D flip-flop for each combination of present state and next state values. The following table shows the characteristic table of T flip-flop along with the **excitation input** of D flip-flop.

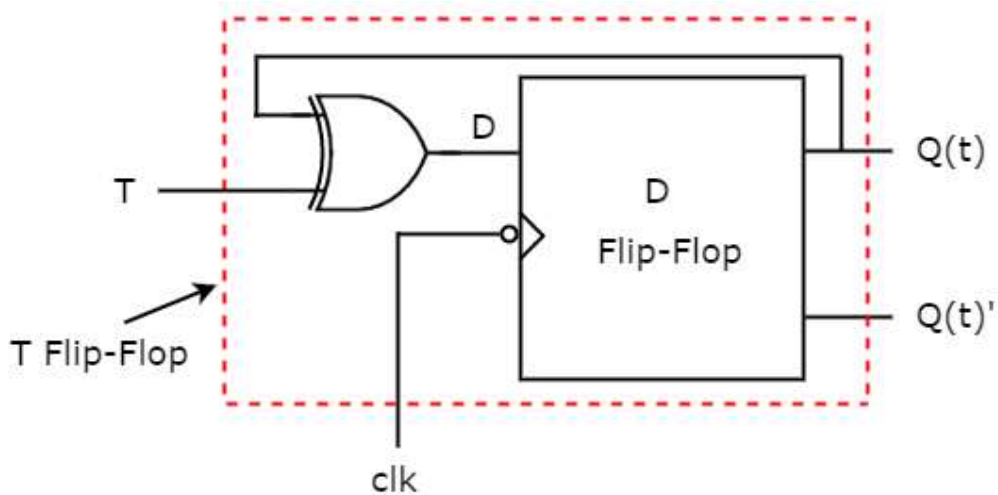
T Flip-Flop Input	Present State	Next State	D Flip-Flop Input
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T	Q(t)	Q(t + 1)	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

From the above table, we can directly write the **Boolean function** of D as below.

$$D = T \oplus Q(t)$$

So, we require a two input Exclusive-OR gate along with D flip-flop. The **circuit diagram** of T flip-flop is shown in the following figure.



This circuit consists of D flip-flop and an Exclusive-OR gate. This Exclusive-OR gate produces an output, which is Ex-OR of T and Q(t). So, the overall circuit has single input, T and two outputs Q(t) & Q(t)'. Hence, it is a **T flip-flop**. Similarly, you can do other two conversions.

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JK Flip-Flop to other Flip-Flop Conversions

Following are the three possible conversions of JK flip-flop to other flip-flops.

- JK Flip-Flop to T Flip-Flop
- JK Flip-Flop to D Flip-Flop
- JK Flip-Flop to SR Flip-Flop

JK Flip-Flop to T Flip-Flop conversion

Here, the given flip-flop is JK flip-flop and the desired flip-flop is T flip-flop. Therefore, consider the following **characteristic table** of T flip-flop.

T Flip-Flop Input	Present State	Next State
T	Q(t)	Q(t + 1)
0	0	0
0	1	1
1	0	1
1	1	0

We know that JK flip-flop has two inputs J & K. So, write down the excitation values of JK flip-flop for each combination of present state and next state values. The following table shows the characteristic table of T flip-flop along with the **excitation inputs** of JK flipflop.

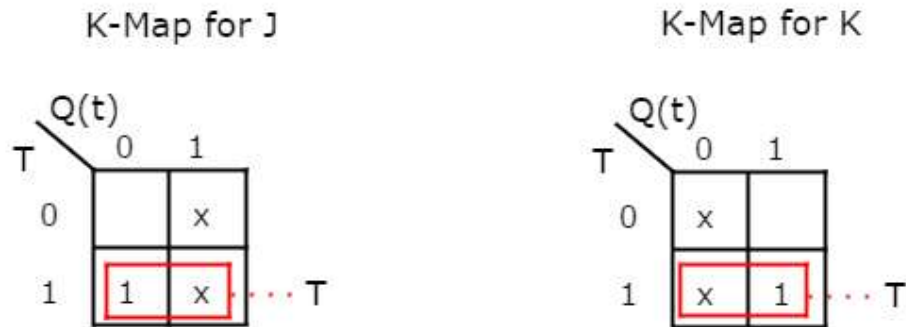
T Flip-Flop Input	Present State	Next State	JK Flip-Flop Inputs	
T	Q(t)	Q(t + 1)	J	K
0	0	0	0	x
0	1	1	x	0
1	0	1	1	x
1	1	0	x	1

From the above table, we can write the **Boolean functions** for each input as below.

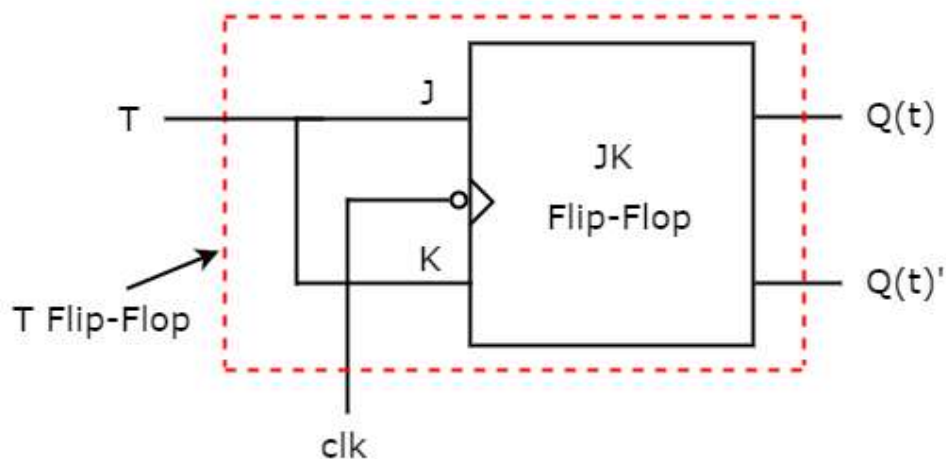
$$J = m_2 + d_1 + d_3$$

$$K = m_3 + d_0 + d_2$$

We can use 2 variable K-Maps for getting simplified expressions for these two inputs. The **k-Maps** for J & K are shown below.



So, we got, $J = T$ & $K = T$ after simplifying. The **circuit diagram** of T flip-flop is shown in the following figure.



This circuit consists of JK flip-flop only. It doesn't require any other gates. Just connect the same input T to both J & K. So, the overall circuit has single input, T and two outputs $Q(t)$ & $Q(t)'$. Hence, it is a **T flip-flop**. Similarly, you can do other two conversions.

T Flip-Flop to other Flip-Flop Conversions

Following are the three possible conversions of T flip-flop to other flip-flops.

- T Flip-Flop to D Flip-Flop
- T Flip-Flop to SR Flip-Flop
- T Flip-Flop to JK Flip-Flop

T Flip-Flop to D Flip-Flop conversion

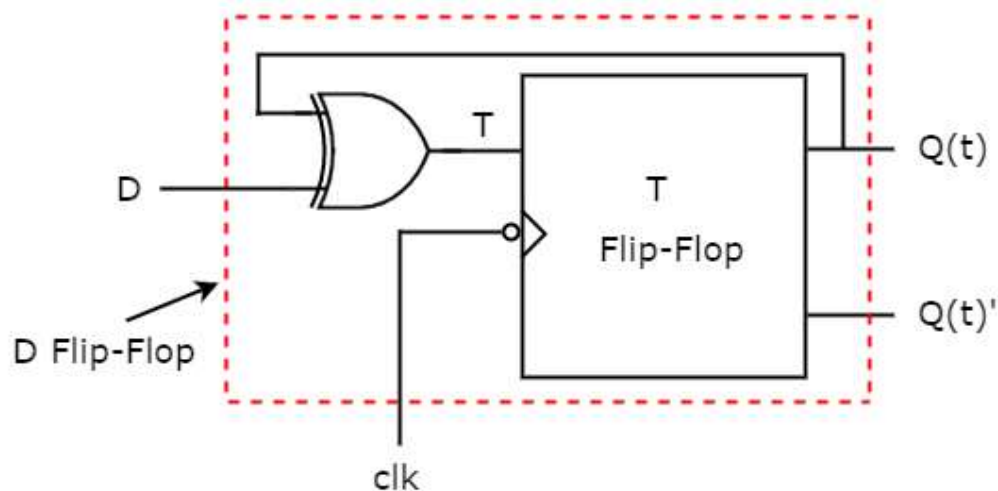
Here, the given flip-flop is T flip-flop and the desired flip-flop is D flip-flop. Therefore, consider the characteristic table of D flip-flop and write down the excitation values of T flip-flop for each combination of present state and next state values. The following table shows the **characteristic table** of D flip-flop along with the **excitation input** of T flip-flop.

D Flip-Flop Input	Present State	Next State	T Flip-Flop Input
D	Q(t)	Q(t + 1)	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

From the above table, we can directly write the Boolean function of T as below.

$$T = D \oplus Q(t)$$

So, we require a two input Exclusive-OR gate along with T flip-flop. The **circuit diagram** of D flip-flop is shown in the following figure.



This circuit consists of T flip-flop and an Exclusive-OR gate. This Exclusive-OR gate produces an output, which is Ex-OR of D and Q(t). So, the overall circuit has single input,

D and two outputs $Q(t)$ & $Q(t)'$. Hence, it is a **D flip-flop**. Similarly, you can do other two conversions.